

$^{35}\text{Cl}(\text{d},\text{t})$ [1981Bh04](#)

$^{35}\text{Cl} \rightarrow 1\text{n} + ^{34}\text{Cl}$ from $J^\pi = 3/2^+$ ^{35}Cl ground state.

[1981Bh04](#): a 23.35-MeV deuteron beam was produced from the Argonne National Laboratory Cyclotron. Targets were prepared by evaporating lead chloride enriched in ^{35}Cl onto thin carbon films. Tritons were detected using a ΔE -E silicon detector telescope with FWHM=140 keV. Measured $\sigma(E_t, \theta)$. Deduced levels, L-transfers, J, π , and spectroscopic factors from DWUCK-DWBA analysis.

 ^{34}Cl Levels

Spectroscopic factor $C^2S = \sigma(\theta)_{\text{exp}} / \sigma(\theta)_{\text{DWBA}} / N$, where $N=3.33$ is a normalization factor adopted from [1966Ba54](#).

<u>E(level)[†]</u>	<u>L[‡]</u>	<u>C²S[‡]</u>
0.0	2	0.41
150	2	1.18
450	0+2	0.11+0.21
670	0+2	0.06+0.22
1210	0+2	0.23+0.09
1840	0+2	0.04+0.12

[†] From [1981Bh04](#), which did not give uncertainties.

[‡] From DWBA analysis of the measured $\sigma(\theta)$ in [1981Bh04](#).